

Forecasting Critical Capability Thresholds in Agents

Project Proposal

1. Introduction

Large Language Models (LLMs) are increasingly deployed as autonomous agents performing long-horizon tasks across domains such as software engineering, cyber operations, and AI research. These deployments carry risks beyond benchmark performance, including harmful behaviors like collusion (Mathew et al., 2024), deceptive reporting (Lermen et al., 2025), and scheming (Carlsmith, 2023). The key evaluation challenge is forecasting when agents will cross capability thresholds, thereby achieving competence on safety-relevant tasks, to enable proactive governance, mitigation, and strategic planning. Recent surveys (Yehudai et al., 2025) highlight difficulties in defining evaluation metrics and benchmarks, while forecasting-oriented studies (Pimpale et al., 2025) propose structured approaches to anticipate performance scaling. This project builds on these foundations to design a tractable framework for forecasting agent capabilities.

2. Target Variables

To make capability forecasting tractable, we focus on target variables that are *measurable* and directly *linked to both performance gains and safety risks*. Prior work (Mohammadi et al., 2025) highlights dimensions such as task efficiency and tool-use accuracy. Building on this, the project considers three main framings:

- *Binary threshold crossing*: forecasting when an agent's probability of success on a benchmark surpasses a predefined threshold (e.g., 70% on SWE-bench, 60% exploit success in cyber tasks). This provides a clear operational definition of capability tied to concrete benchmarks.
- *Time-to-threshold crossing*: estimating the scale or calendar time at which the above threshold is crossed. This shifts the focus from *whether* an event will occur to *when*, which is crucial for planning.
- *Emergence of harmful behaviors*: predicting when unsafe behaviors such as collusion, deception, or escalation first appear in simulated or adversarial environments. These are safety-critical and must be modelled explicitly as binary events.

The proposal prioritises *binary threshold crossing* as it is tractable, measurable, and directly aligned with agent performance. Approach C extends this to unsafe behavior detection by treating harmful outcomes as binary events in adversarial trials.

3. Comparison with Benchmark Forecasting

Forecasting agent capabilities shares similarities with benchmark forecasting but also introduces additional challenges:

- *Sequential decision-making*: Agent tasks involve multiple interdependent steps, where early errors compound into failure.
- *External interactions*: Agents use tools, APIs, or environments, adding stochasticity and variance to outcomes.
- *Unsafe behaviors*: Capabilities must be forecasted alongside harmful behaviors such as collusion or deception, which are absent from accuracy metrics.
- *Ground truth*: Long-horizon tasks may admit multiple valid trajectories or partial success, making outcomes harder to define than in single-turn QA benchmarks.

4. Forecasting Approaches

Approach A. Scaling-law extrapolation with Bayesian regression

This approach proposes to forecast capability thresholds by fitting scaling laws to existing benchmark data. Model performance is often found to follow predictable trends with respect to size, compute, or data (Kaplan et al., 2020). A simple functional form is a power law: $y = \alpha \cdot (\text{scale})^\beta$

where, y represents performance on a benchmark, *scale* captures model size or compute, and α, β are fitted parameters. Bayesian regression extends this by placing priors on the parameters and estimating posterior distributions, which allows forecasts to include credible intervals instead of point estimates.

The method can be reframed to focus directly on threshold events, such as predicting when a model family will exceed 70% success on SWE-bench. Rather than modelling *average* performance trends, the emphasis is on *when the probability of success first passes a predefined boundary*. Recent work by Pimpale et al. (2025) has applied this framing to multi-step agent tasks, while Ruan et al. (2024) considered scaling laws for emergent phenomena.

Its main *advantages* are *simplicity*, *interpretability*, and *modest data requirements*, while providing uncertainty estimates. However, scaling laws *assume smooth progress* and often fail to capture discontinuous jumps or emergent behaviors, and they depend entirely on *sparse historical data*, especially for long-horizon or safety-critical tasks.

Approach B. Survival / Time-to-Event Modelling

This approach frames capability forecasting as a *time-to-event problem*, where the “event” is the crossing of a threshold such as achieving 70% success on SWE-bench or the first observed instance of a harmful behavior. The proposal is to leverage a proportional hazards model (Cox, 1972) that estimates the instantaneous probability of the event as: $h(t|x) = h_0(t) \cdot \exp(\beta^T x)$

where, $h_0(t)$ is the baseline hazard over time or scale, while the covariates x (for example, model scale, dataset size, architecture, or fine-tuning strategy) determine how quickly the threshold is reached. A central advantage is that censored data, where agents have not yet crossed the threshold, still contribute to the analysis. The coefficients β are interpretable in terms of hazard ratios. For example, if doubling dataset size corresponds to a hazard ratio of 1.5, the threshold is 50 percent more likely to be crossed at any given scale.

Compared with scaling-law extrapolation, survival models can incorporate multiple explanatory factors and produce interpretable forecasts. The main limitations are the requirement for longitudinal

data across model generations, which is often scarce and proprietary, and the difficulty of capturing rare harmful behaviors unless adversarial evaluation data is explicitly collected.

Approach C. Simulation-Based Benchmarking with Hierarchical Bayesian Modelling

This approach proposes to *measure when agents cross capability thresholds* by running them in controlled simulations and recording *binary outcomes*, such as whether success exceeds 70% on SWE-bench or when the probability of an unsafe behavior (e.g., collusion, deception) exceeds 10% in adversarial scenarios. *Both safe and unsafe outcomes can be treated as binary events.*

Unlike Approach A, which relies only on historical benchmarks, this method actively *generates new data through repeated trials*. The probability of success for an agent i on task j can be expressed as:

$$\text{logit}(p_{ij}) = \alpha_j + \beta \cdot \log(\text{params}_i) + u_i$$

Here, α_j represents task difficulty, β captures the effect of model scale, and u_i accounts for variation across agents. Bayesian inference over these parameters will produce forecasts with quantified uncertainty, for example estimating the probability that a model family will exceed a given threshold at its next scale-up. When simulator likelihoods are too complex to specify, Approximate Bayesian Computation (Toni et al., 2009) can be used to approximate posterior distributions, i.e. simulate outcomes under candidate parameter settings, compare summary statistics with observed data, and accept parameter draws that closely match.

Existing benchmarks like SafeArena (Zhou et al., 2025) use simulation-based evaluation to measure safety properties of autonomous web agents, but they are descriptive and do not forecast future threshold crossings. The proposed approach extends this by applying Bayesian inference to simulation outcomes, predicting when agents are likely to exceed capability or safety thresholds. However, simulation-based methods are computationally intensive, requiring hundreds of trials per configuration ($\approx 200\text{--}500$), and may suffer from a *simulation–reality gap*. *Unknown unknowns* remain, as human designers cannot anticipate all behaviors. Modeling harmful behaviors such as collusion would require resource-intensive adversarial task design.

5. Recommended Method and Project Plan

Approach C is recommended because it directly measures task success and unsafe behaviors under controlled simulation, rather than relying on sparse historical data or assuming smooth capability growth. The hierarchical Bayesian framework provides interpretable, probabilistic forecasts that are directly relevant for governance and safety planning. Unlike Approaches A and B, which extrapolate from past data, Approach C allows targeted stress-testing in domains and adversarial scenarios where harmful behaviors are more likely to emerge. Although resource-intensive, this flexibility makes it the most reliable and actionable method for anticipating safety-relevant thresholds.

Project structure and Progress (2 months)

- *Weeks 1–2:* Define two domains (e.g., SWE and cyber) and specify thresholds ($\geq 70\%$ task success, $\geq 50\%$ exploit success). *Deliverable:* evaluation plan.
- *Weeks 2–4:* Build lightweight simulators and run pilot trials (50–100 per configuration). *Deliverable:* dataset schema and preliminary outcomes.

- *Weeks 5–7*: Scale up to 200–400 trials per configuration. Fit Bayesian models or ABC approximations. *Deliverable*: draft forecasts with uncertainty intervals.
- *Week 8*: Back-test against past performance, sensitivity analysis, and final report. *Deliverable*: final report and success probability curves vs scale.

Resources

- *Data*: SWE-bench, cyber CTF-style environments, adversarial task variants.
- *Compute*: GPUs for open-source models; API budget for GPT-class baselines.
- *Personnel*: One ML researcher, one domain expert, one safety evaluator.

Testing/Validation

- *Posterior predictive checks* to confirm that the fitted Bayesian models reproduce observed simulation outcomes.
- *Back-testing* against prior model generations to test whether forecasts made retrospectively would have matched observed performance.
- *Cross-validation across domains and tasks* to assess generalisability of results.
- *Sensitivity analysis* on priors, simulation assumptions, and adversarial task design to test stability of conclusions.

6. Conclusion

This proposal suggests forecasting approaches for agent capabilities and recommends simulation-based benchmarking with hierarchical Bayesian modelling. While resource-intensive, it best captures both task success and unsafe behaviors, providing the most reliable basis for proactive governance.

References

- Carlsmith, J. (2023). *Scheming AIs: Will AIs fake alignment during training in order to get power?* arXiv:2311.08379. <https://arxiv.org/abs/2311.08379>
- Kaplan, J., McCandlish, S., Henighan, T., et al. (2020). *Scaling laws for neural language models*. arXiv:2001.08361. <https://arxiv.org/abs/2001.08361>
- Lermen, C., Dziemian, T., Antolín, P., et al. (2025). *Deceptive automated interpretability: Language models coordinating to fool oversight systems*. arXiv:2504.07831. <https://arxiv.org/abs/2504.07831>
- Mathew, A., Lermen, C., Clark, J., et al. (2024). *Hidden in plain text: Emergence and mitigation of steganographic collusion in LLMs*. arXiv:2410.03768. <https://arxiv.org/abs/2410.03768>
- Mohammadi, S., Zhang, K., Perez, E., et al. (2025). *Agent capability and behaviour evaluation: Criteria and frameworks*.
- Pimpale, A., Shah, R., Gomez, L., et al. (2025). *Forecasting frontier language model agent capabilities*. arXiv:2502.15850. <https://arxiv.org/abs/2502.15850>
- Ruan, Y., Gao, L., Narayanan, A., et al. (2024). *Observational scaling laws and the predictability of language models*. NeurIPS 2024.
- Yehudai, A., Eden, L., Li, A., Uziel, G., Zhao, Y., Bar-Haim, R., Cohan, A., & Shmueli-Scheuer, M. (2025). *Survey on Evaluation of LLM-based Agents*. arXiv:2503.16416. <https://arxiv.org/abs/2503.16416>
- Toni, T., Welch, D., Strelkowa, N., Ipsen, A., & Stumpf, M. P. H. (2009). *Approximate Bayesian computation scheme for parameter inference and model selection in dynamical systems*. arXiv:0907.0374. <https://arxiv.org/abs/0907.0374>

- Zhou, R., Wang, J., Lin, Z., et al. (2025). *SafeArena: Evaluating the safety of autonomous web agents*. arXiv:2503.04957. <https://arxiv.org/abs/2503.04957>
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2), 187–202.